QPCODE **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) EXAMINATION

III SEMESTER

DSE: MG3DSEPHY202 INTRODUCTION TO SPACE PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1.5 hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions Answer any

10 questions

Each question carries **2** marks

1.	Mention two situations of naturally occurring plasma is found.	[K] [CO1]
2.	State the relationship between electric field E and magnetic field B in an electromagnetic wave.	[U] [CO1]
3.	Give an account on the Karman line and its significance.	[Ap] [CO6]
4.	What is meant by a geomagnetic storm.	[I] [CO6]
5.	Name the layers of Solar atmosphere.	[K] [CO2]
6.	Distinguish between solar flare and solar prominence.	[S] [CO5]
7.	What is meant by the term space weather?	[U] [CO2]
8.	Name any two space weather models with examples.	[E] [CO2]
9.	Which unique position in space does Aditya-L1 occupy, and why is it important?	[A] [CO4]
10.	Comment on Earth's magnetic field components.	[K] [CO3]
11.	Explain the structure of Earth's magnetosphere.	[U] [CO3]
12.	Discuss the formation of Earth's ionosphere.	[An] [CO4]
13.	Explain the propagation of radio waves through ionosphere and obtain the equation for critical frequency.	[A] [CO4]
14.	Comment on the measurement of TEC using GPS data.	[An] [CO5]

[2x10 = 20]

Part B

Short Essay Type Questions
Answer any **6** questions
Each question carries **5** marks

15. Briefly outline the significance of space physics. [K] [CO1]
16. Derive and explain the rocket equation. [An] [CO6]
17. With a neat diagram explain the internal structure of Sun. [K] [CO2]
18. Explain the space weather Toy model by discussing the three phases of a geomagnetic substorm. [U] [CO2]
19. Briefly explain Indias Solar Mission Aditya L1. Compare its specialities with Nasa's Parker Solar Probe. [A] [CO4]
20. Explain the interaction of planetary atmosphere with solar wind. [U] [CO3]
21. Discuss the structure of Earth's ionosphere in detail. [An][CO4]
22. Briefly explain the measurement of total electron content (TEC) using ionosonde. [An] [CO5]

[5x6 = 30]